LESSON 4.5
EXPLORING RELATIONSHIPS IN CIRCLES

LESSON INTRODUCTION

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can explain the relationship between the circumference and diameter of a circle.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- What is the relationship between the diameter and circumference of a circle?

LESSON NARRATIVE

Students explore the relationship between the diameter and circumference of a circle using a string, ruler, and circular objects. Students create a conjecture about the relationship, discovering the proportional relationship between circumference and diameter as well as the function of pi in that proportional relationship.

COMPONENT SUMMARY

4.5.1 WARM-UP *(~5 MIN.)*

Convert ounces to pounds to write ratios of hummingbirds to ostriches

4.5.2 GUIDED INSTRUCTION *(10 MIN.)*

Review key terms

4.5.3 EXPLORATION *(25 MIN.)*

Explore relationship between the diameter and circumference of a circle

4.5.4 WRAP-UP *(~5 MIN.)*

Explain the relationship between circumference of circle and its diameter in a letter to an absent student

RESOURCES

GLOSSARY

Key Terms:

- Circle
- Radius
- Diameter
- Circumference

Additional Terms:

- N/A

MATERIALS

For each group:

- 5 feet of string
- Rulers
- 5 Circular items
- Cardboard circles
- Calculator

ADDITIONAL PREPARATION

- Prepare five different-sized circular items. You could use items such as lids to various-sized containers. Consider marking the centers and labeling the circular items A, B, C, D, and E.
- Prepare a graph using your measurements for the objects. If time is an issue in Exploration 4.5.3, then have students use your graph to answer questions 4 and 5.
- For question 9 in 4.5.3 Exploration, prepare cardboard circles with the following diameters: 1 foot, 6.25 inches, and 8.5 centimeters.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not realize that the diameter can be drawn from any two points on opposite sides of a circle that go through the center.
- Students may not realize that the circumference is a length.

THINGS TO CONSIDER

- In 4.5.3 Exploration, if students are struggling to recognize the relationship between the diameter and circumference, prompt them to check that they measured correctly.
- Consider allowing students to have a calculator.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 4.5.3 Exploration, ask students to pause after reading each question and reflect on what they notice and wonder. Then, ask them to discuss what they noticed and wondered with a partner before sharing with the whole class.

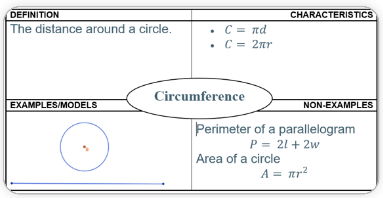
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 4.5.3 Exploration, students explore, discover, and represent the proportional relationships between diameter and circumference of circles in tables and graphs. Students develop an understanding of the relationship by looking for and making use of structure.

DIFFERENTIATE INSTRUCTION

Utilize vocabulary strategies for students to formulate working definitions of key words while constructing natural scaffolding for use in current and future lessons. A Frayer Model provides students with a visual graphic organizer to leverage for words like radius, diameter, and circumference.



4.5.1 WARM-UP: OSTRICH VS HUMMINGBIRD

TEACHER MOVES

Students often struggle comparing values with different units. The focus of this warm-up for the student should be number sense and grasping the size difference between the two birds. Students may be surprised by how little a hummingbird weighs and how heavy an ostrich is.

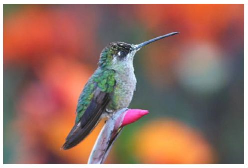
ENRICHMENT

Are the average weights given in the same unit? How can you convert the average weight of a hummingbird to pounds? Can you express $\frac{0.15}{16}$ lbs. as a decimal?

4.5.1 Warm-Up: Ostrich vs Hummingbird

The average weight of a hummingbird is 0.15 ounces. The average weight of an ostrich is 230 pounds.

1. What is the ratio of the average weight of an ostrich to the average weight of a hummingbird?
   1 pound $=$ 16 ounces.



Use the fact that 1 pound $=$ 16 ounces to find the weight of a hummingbird in pounds:

$$\frac{1lb}{16oz} \times \frac{0.15oz}{1} = 0.009lb$$

The ratio of ostrich to hummingbird in pounds is $\frac{230}{0.009}$

Since the question did not specify to find the ratio in pounds or ounces, this problem could also be solved by finding the weight of the ostrich in ounces and then creating the ratio in ounces.

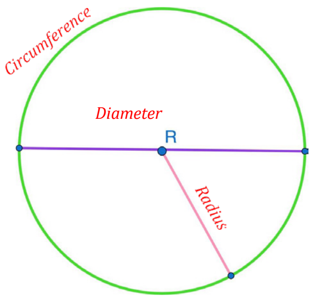
$$\frac{16\ oz}{1\ lb} \times \frac{230\ lb}{1} = 3680\ oz$$

The ratio of ostrich to hummingbird in ounces is $\frac{3680}{0.15}$

4.5.2 Guided Instruction: Identifying Parts of a Circle

A ***circle*** is a perfectly round two-dimensional figure, where all points on the circle are equidistant from the center.

A circle with center R is shown.



The purple line segment is a diameter of the circle. The pink line segment is the radius of the circle. The center of a circle is the point that is the same distance from all points on the circle.

The plural of radius is radii. Circles have radii and diameters, both of which pass through the center of the circle.

The green part of the circle is the circumference.

1. Label each part of circle R by writing the name of each part.

A ***radius*** is a line segment extending from the center of a circle to a point on the circle.

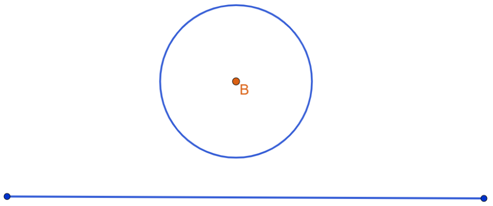
A ***diameter*** is a line segment from any point on the circle passing through the center to another point on the circle.

The length of a diameter is equal to the length of 2 radii.

The ***circumference*** is the distance around a circle.

If a circle were made of a piece of string and we cut the string and straightened it out, the circumference would be the length of the string.

Circle B is shown.



The line segment is approximately the same length as the circumference of circle B.

4.5.2 GUIDED INSTRUCTION: IDENTIFYING PARTS OF A CIRCLE

ENRICHMENT

What do you notice about the relationship between the radius and diameter? Draw a different diameter and radius on the diagram. Does your diameter go through the center? Does your radius touch the center? What do you wonder about circles?

DIFFERENTIATE INSTRUCTION

Consider providing kinesthetic and visual entry points by prompting students to use a ruler to measure the diameter and the radius of the circle.

DIFFERENTIATE INSTRUCTION

!!! Utilize vocabulary strategies like a Frayer Model or a word wall to give students opportunities to engage in language acquisition development.

TEACHER MOVES

Take the time to illustrate for the students how a circumference can be laid out into a straight line. This will help students recognize that the units for length are not square units. If you have a flexible tape measure it may be helpful to demonstrate measuring the circumference of a circular object and then laying the tape measure flat.

4.5.3 EXPLORATION: FINDING CIRCUMFERENCE

DIFFERENTIATE INSTRUCTION

Having students work in groups will enable them to share supplies and discuss their methods and answers.

Provide each group with five different-sized circular items. You could use items such as lids of various sized containers.

Consider marking the centers and labeling the circular items A, B, C, D, and E.

TEACHER MOVES

The values in the key are possible values. Students are not expected to have exact answers. Students should measure the circumference to the nearest mm. It may be necessary to demonstrate to students how to read the values. Tell students to be as precise as possible.

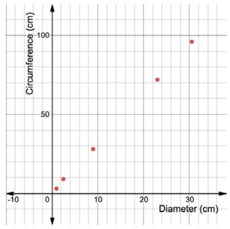
4.5.3 Exploration: Finding Circumference

1. In the table, record the name of the five different circular objects your group is using.
2. Using a ruler, measure and record the diameter of each circle in centimeters (cm).
3. Using string, wrap the circular object with string using your finger to mark the circumference of the circle. Straighten out the string and use it to measure the length of the circumference in centimeters (cm) with a ruler. Then, record it in the table. *Answers will vary.*

Object	Diameter	Circumference
Glue Stick	1	~3
Water Bottle	2.5	~7.8
Plate	9	~28
Trash Can	23	~72
Clock	30.5	~96

A coordinate plane is shown with the horizontal axis representing the diameter and the vertical axis representing the circumference.

4. Plot the values for each object on the graph. *Answers will vary*



5. Discuss with your partner what you notice about the graph.
   Answers will vary. I noticed that there is a proportional relationship. I notice that the circumference is about 3 times bigger than the diameter.

6. Using circumference (C) and diameter (D), find the value of the expressions listed in the table for each of the circular objects measured.

Object	$C+D$	$C-D$	$C \times D$	$C \div D$
Glue Stick	4	2	3	3.0
Water Bottle	10.3	5.3	29.5	3.12
Plate	37	19	252	3.11
Trash Can	95	49	1,656	3.13
Clock	126.5	60.5	2,928	3.15

7. What do you notice about the table?
   Answers will vary. I notice that the last column is the only column that has similar values. I notice that the last numbers in the last column equal the value of pi.

8. Use the information in the table to complete the statement.

   The value of
   - ○ $C+D$
   - ○ $C-D$
   - ○ $C \times D$
   - ○ $C \div D$

   is approximately equivalent for each circular object. It is approximately 3.14.

4.5.3 EXPLORATION: FINDING CIRCUMFERENCE (CONTINUED)

TEACHER MOVES

Notice and Wonder

Ask students to pause after plotting the values and reflect on what they notice and wonder. Then, ask them to discuss what they noticed and wondered with a partner before sharing with the whole class.

TEACHER MOVES

Because student measurements are not precise, their graphs may not show a clear proportional relationship. If students are unsure, then ask them to consider how to best describe the graph. Whether a student indicates the relationship on the graph is proportional or demonstrates that the points seem to align is not the purpose of this component. Rather, this lays the groundwork for future lessons on proportional relationships.

TEACHER MOVES

If students are struggling to recognize the relationship between the diameter and circumference, prompt them to check that they measured correctly.

4.5.3 EXPLORATION: FINDING CIRCUMFERENCE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing groups with cardboard circles with the following diameters: 1 foot diameter, 6.25 inches diameter, and $8\frac{1}{2}$ centimeters.

9. Use the relationship you observed to predict the circumference of each circle using the given diameter. Record your predictions in the middle column.

Given Diameter	Predicted Circumference	Measured Circumference
1 foot (ft)	3(ft)	3.1 (ft)
6.25 inches (in.)	18.75 (in.)	19.5(in.)
$8\frac{1}{2}$ centimeters (cm)	25.5 (cm)	26.5 (cm)

10. Use a string and a ruler to measure the circumference of each of the circles using the same units from the given diameter. Then, record the measured circumference of each circle in the table.
See above in the table.

11. Describe the accuracy of your predictions.
Answers will vary. My predictions were not accurate but close. I predicted that each circle's circumference would be 3 times the given diameter.

4.5.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about the relationship between the circumference of a circle and its diameter.

   Answers will vary. We learned that a circle's circumference is about 3.14 *times its diameter.*

4.5.4 WRAP-UP: COOL DOWN

TEACHER MOVES

Consider providing the following sentence stems:

- The relationship between the diameter of a circle and its circumference is…
- Pi represents…
- There is a ________ relationship between diameter and circumference, and it is….